Linear Regression

Modelo: $p(y|\mathbf{x},\mathbf{w}) = \mathcal{N}(y|\mathbf{w}^\top\mathbf{x},\sigma^2)$
Predicción: $\hat{y} = \mathbb{E}[y|\mathbf{x},\mathbf{w}] = \mathbf{w}^\top\mathbf{x}$
$\mathcal{L} = \text{MSE} = \frac{1}{N}\sum_i (y_i - \mathbf{w}^\top\mathbf{x}_i)^2$
$\nabla_\mathbf{w}\mathcal{L} = \frac{2}{N}\sum_i(\mathbf{w}^\top\mathbf{x}_i - y_i)\mathbf{x}_i = \frac{2}{N}\mathbf{X}^\top(\mathbf{X}\mathbf{w}-\mathbf{y})$
Solución (MLE / LS): $\mathbf{w}^* = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}$

Logistic Regression

Binaria
Modelo: $p(y|\mathbf{x},\boldsymbol{\theta}) = \text{Ber}(y|\sigma(\mathbf{w}^\top\mathbf{x}+b))$
Predicción (prob): $\mu_i = \sigma(\mathbf{w}^\top\mathbf{x}_i)$
$\mathcal{L} = \frac{1}{N}\sum_i \mathbb{H}_{ce}(y_i,\mu_i)$
$\nabla_\mathbf{w}\mathcal{L} = \frac{1}{N}\sum_i(\mu_i - y_i)\mathbf{x}_i = \frac{1}{N}\mathbf{X}^\top(\boldsymbol{\mu}-\mathbf{y})$
Multiclase
Modelo: $p(y|\mathbf{x},\mathbf{w}) = \text{Cat}(y|\text{Softmax}(\mathbf{w}^\top\mathbf{x}))$
Predicción (prob): $\boldsymbol{\mu}_i = \text{Softmax}(\mathbf{w}^\top\mathbf{x}_i)$
$\mathcal{L} = \frac{1}{N}\sum_i \mathbb{H}_{ce}(\mathbf{y}_i,\boldsymbol{\mu}_i)$
$\nabla_\mathbf{w}\mathcal{L} = \frac{1}{N}\sum_i \mathbf{x}_i(\boldsymbol{\mu}_i - \mathbf{y}_i)^\top = \frac{1}{N}\mathbf{X}^\top(\mathbf{M}-\mathbf{Y})$
($\mathbf{M}$ matriz de predicciones, $\mathbf{Y}$ one-hots)

GDA

Modelo: $p(y=c|\mathbf{x},\mathbf{w}) \propto \pi_c p(\mathbf{x}|y=c,\mathbf{w})$
Class-cond: $p(\mathbf{x}|y=c,\mathbf{w}) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_c,\boldsymbol{\Sigma}_c)$
Prior: $\pi_c = p(y=c|\mathbf{w})$
Entrenamiento: MLE $(\hat{\pi},\hat{\boldsymbol{\mu}}_c,\hat{\boldsymbol{\Sigma}}_c)$
MAP: $\hat{\boldsymbol{\Sigma}}_{\text{map}} = \lambda\text{diag}(\hat{\boldsymbol{\Sigma}}) + (1-\lambda)\hat{\boldsymbol{\Sigma}}$
QDA ($\boldsymbol{\Sigma}_c$ libre por clase)
Frontera: $\mathbf{x}^\top\mathbf{A}\mathbf{x} + \mathbf{b}^\top\mathbf{x} + c_0 = 0$
$\mathbf{A} = \frac{1}{2}(\boldsymbol{\Sigma}_{c'}^{-1} - \boldsymbol{\Sigma}_c^{-1})$, $\mathbf{b} = \boldsymbol{\Sigma}_c^{-1}\boldsymbol{\mu}_c - \boldsymbol{\Sigma}_{c'}^{-1}\boldsymbol{\mu}_{c'}$
$c_0 = \log\frac{\pi_c}{\pi_{c'}} - \frac{1}{2}\log\frac{|\boldsymbol{\Sigma}_c|}{|\boldsymbol{\Sigma}_{c'}|} - \frac{1}{2}(\boldsymbol{\mu}_c^\top\boldsymbol{\Sigma}_c^{-1}\boldsymbol{\mu}_c - \boldsymbol{\mu}_{c'}^\top\boldsymbol{\Sigma}_{c'}^{-1}\boldsymbol{\mu}_{c'})$
LDA ($\boldsymbol{\Sigma}_c = \boldsymbol{\Sigma}\ \forall c$)
Frontera: $\mathbf{w}^\top\mathbf{x} + w_0 = 0$
donde $\mathbf{w} = \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_c - \boldsymbol{\mu}_{c'})$
$w_0 = \log\frac{\pi_c}{\pi_{c'}} - \frac{1}{2}(\boldsymbol{\mu}_c + \boldsymbol{\mu}_{c'})^\top\mathbf{w}$

Naive Bayes

Naive Bayes Assumption: $x_d \perp x_{d'} \mid y = c$
Modelo: $p(y=c|\mathbf{x}) \propto \pi_c \prod_{d=1}^D p(x_d|y=c,\boldsymbol{\theta}_{dc})$
$p(x_d|y=c,\boldsymbol{\theta}_{dc})$ según tipo de x_d:
- Binaria $\text{Ber}(x_d|\theta_{dc})$
- Categórica $\text{Cat}(x_d|\boldsymbol{\theta}_{dc})$
- Real $\mathcal{N}(x_d|\mu_{dc},\sigma^2_{dc})$

Entrenamiento: MLE

CART

$\mathcal{L}(\boldsymbol{\theta}) = \sum_i \ell(y_i, f(\mathbf{x}_i;\boldsymbol{\theta}))$
Split (nodo i):
$\arg\min_j \min_t \frac{|\mathcal{D}_i^L|}{|\mathcal{D}_i|}C^L(\mathcal{D}_i^L) + \frac{|\mathcal{D}_i^R|}{|\mathcal{D}_i|}C^R(\mathcal{D}_i^R)$
Regresión: $C(\mathcal{D}_i) = \text{MSE}$
Clasificación: $C(\mathcal{D}_i) = G_i = \sum_c \hat{\pi}_{ic}(1-\hat{\pi}_{ic})$ ó $C(\mathcal{D}_i) = H_i = \mathbb{H}(\hat{\boldsymbol{\pi}}_i) = -\sum_{ic}\hat{\pi}_{ic}\log(\hat{\pi}_{ic})$

Multi-layer Perceptron (MLP)

$\mathbf{z}^{(\ell)} \in \mathbb{R}^{M^{(\ell)}}$: output capa ℓ
$\mathbf{a}^{(\ell)} \in \mathbb{R}^{M^{(\ell)}}$: preactivación capa ℓ
$\boldsymbol{\delta}^{(\ell)} \in \mathbb{R}^{M^{(\ell)}}$: error de propagación en capa ℓ
$\mathbf{W}^{(\ell)} \in \mathbb{R}^{M^{(\ell)}\times M^{(\ell-1)}}$: params entre ℓ y $\ell-1$
Back-propagation

 # Forward-pass
$\mathbf{z}^{(0)} \leftarrow \mathbf{x}_i$
for $\ell = 1:L$ **do**
  $\mathbf{a}^{(\ell)} \leftarrow \mathbf{W}^{(\ell)}\mathbf{z}^{(\ell-1)} + \mathbf{b}^{(\ell)}$
  $\mathbf{z}^{(\ell)} \leftarrow \varphi^{(\ell)}(\mathbf{a}^{(\ell)})$
$\hat{\mathbf{y}}_i \leftarrow \mathbf{z}^{(L)}$, $\mathcal{L}_i \leftarrow \mathcal{L}_i(\mathbf{y}_i,\hat{\mathbf{y}}_i)$
 # Backward-pass
$\delta^{(L)} \leftarrow \varphi'^{(L)}(\mathbf{a}^{(L)}) \odot \frac{\partial\mathcal{L}_i}{\partial\hat{\mathbf{y}}_i}$
$\frac{\partial\mathcal{L}_i}{\partial\mathbf{W}^{(L)}} \leftarrow \delta^{(L)}\mathbf{z}^{(L-1)\top}$
for $l = L-1:1$ **do**
  $\delta^{(\ell)} \leftarrow \varphi'^{(\ell)}(\mathbf{a}^{(\ell)}) \odot \mathbf{W}^{(\ell+1)\top}\delta^{(\ell+1)}$
  $\frac{\partial\mathcal{L}_i}{\partial\mathbf{W}^{(\ell)}} \leftarrow \delta^{(\ell)}\mathbf{z}^{(\ell-1)\top}$
  $\frac{\partial\mathcal{L}_i}{\partial\mathbf{b}^{(l)}} \leftarrow \delta^{(\ell)}$

Non-Parametric Methods

KNN-classifier:
$p(y=c|\mathbf{x},\mathcal{D}) = \frac{1}{K}\sum_{i\in\mathcal{N}_K(\mathbf{x},\mathcal{D})}\mathbb{I}(y_i=c)$
Parzen Window Density Estimator:
$p(\mathbf{x},\mathcal{D}) = \frac{1}{N}\sum_{i=1}^N \mathcal{K}_h(\mathbf{x},\mathbf{x}_i)$

Perceptron

$\hat{y} = \mathbb{I}(\mathbf{w}^\top\mathbf{x}+b>0) \to$ (determinístico)
Update: $\mathbf{w}_{t+1} \leftarrow \mathbf{w}_t - \eta(\hat{y}_i - y_i)\mathbf{x}_i$

Activation Functions (φ)

Sigmoid: $\sigma(z) = \frac{1}{1+e^{-z}}$; $\sigma' = \sigma(1-\sigma)$
Tanh: $\tanh' = 1-\tanh^2$
ReLU: $\max(0,z)$; $\text{ReLU}' = \mathbb{I}[z>0]$
Leaky ReLU: $\max(\alpha z, z)$, $\alpha \ll 1$
Softplus: $\log(1+e^z)$
Identity: $\varphi(z) = z$
Softmax: $\sigma(\mathbf{z})_i = e^{z_i}/\sum_j e^{z_j}$

NN Regularization + ∇ vanishing sol.

Weight Decay (ℓ_2): MAP estimation
$p(\mathbf{w}) = \mathcal{N}(\mathbf{w}|\mathbf{0},\alpha^2\mathbf{I})$, $p(\mathbf{b}) = \mathcal{N}(\mathbf{b}|\mathbf{0},\beta^2\mathbf{I})$
Dropout: $\theta_{lji} = w_{lji}\epsilon_{li}$, $\epsilon \sim \text{Ber}(1-p)$
Al predecir: escalar activaciones por $(1-p)$
Early Stopping: frenar cuando $\uparrow \mathcal{L}_{\text{val}}$
Batch Normalization: normalizar activaciones por mini-batch $\mathcal{B}$
$\hat{z}^{(\ell)} = \frac{z^{(\ell)} - \mu_\mathcal{B}}{\sqrt{\sigma^2_\mathcal{B}+\varepsilon}}$, $\tilde{z}^{(\ell)} = \gamma\hat{z}^{(\ell)} + \beta$
Glorot/Xavier initialization
$w_{ij} \sim \mathcal{N}(0,\sigma^2)$ con $\sigma^2 = 2/(n_{\text{in}}+n_{\text{out}})$
Activaciones sin saturación: ReLU, Leaky ReLU (evitan saturación)
Gradient clipping: $\mathbf{g} = \min\left(0, \mathbf{g}\cdot\frac{c}{\|\mathbf{g}\|}\right)$
Batch Normalization / Layer Normalization

Optimization

GD: $\mathbf{w}_{t+1} \leftarrow \mathbf{w}_t - \eta_t\nabla_\mathbf{w}\mathcal{L}|_{\mathbf{w}_t}$
Newton: $\mathbf{w}_{t+1} \leftarrow \mathbf{w}_t - \eta_t\mathbf{H}_t^{-1}\nabla_\mathbf{w}\mathcal{L}|_{\mathbf{w}_t}$
Momentum: $\begin{cases}\mathbf{w}_{t+1} \leftarrow -\eta_t\mathbf{m}_t\\ \mathbf{m}_t \leftarrow \beta\mathbf{m}_{t-1} + \nabla\mathcal{L}_\mathbf{w}|_{\mathbf{w}_{t-1}}\end{cases}$
Learning-rate Schedule:
- Lineal: $\eta_t = (1-\frac{t}{T})\eta_0 + (\frac{t}{T})\eta_T$
- Potencia: $\eta_t = \eta_0(1+\frac{t}{s})^c$
- Exponencial: $\eta_t = \eta_0\cdot c^{(t/s)}$

AdaGrad: $\begin{cases}w_{t+1,d} \leftarrow w_{t,d} - \eta\frac{1}{\sqrt{s_{k,d}+\epsilon}}\frac{\partial\mathcal{L}}{\partial w_d}\\ s_{t,d} = \sum_{t=1}^{T-1}(\frac{\partial\mathcal{L}_t}{\partial w_d})^2\end{cases}$
ADAM: $\begin{cases}w_{t+1,d} \leftarrow w_{t,d} - \eta\frac{\hat{s}_{t,d}}{\sqrt{\hat{r}_{k,d}+\epsilon}}\\ \hat{s}_{t,d} = s_{t,d}/(1-\beta_{1t})\\ \hat{r}_{t,d} = r_{t,d}/(1-\beta_{2t})\\ s_{t,d} = \beta_1 s_{t-1,d} + (1-\beta_1)\frac{\partial\mathcal{L}}{\partial w_d}\\ r_{t,d} = \beta_2 r_{t-1,d} + (1-\beta_2)(\frac{\partial\mathcal{L}}{\partial w_d})^2\end{cases}$

Regression Metrics

$e_i \triangleq y_i - \hat{y}_i$; $\text{RSS} \triangleq \sum_i e_i^2$; $\text{TSS} \triangleq \sum_i(y_i-\bar{y})^2$
$\text{MSE} = \frac{1}{N}\text{RSS}$
$\text{RMSE} = \sqrt{\text{MSE}}$
$\text{MAE} = \frac{1}{N}\sum_i|e_i|$
$R^2 \triangleq 1 - \frac{\text{RSS}}{\text{TSS}} \in [0,1]$ (varianza explicada)

Classification Metrics

$\text{Acc} = \frac{TP+TN}{TP+TN+FP+FN}$ $\text{Recall} = \frac{TP}{TP+FN}$
$\text{Precision} = \frac{TP}{TP+FP}$ $\text{FPR} = \frac{FP}{FP+TN}$
$\text{FDR} = \frac{FP}{FP+TP}$
$F_1 = \frac{2\cdot\text{Precision}\cdot\text{Recall}}{\text{Precision}+\text{Recall}} = \frac{2TP}{2TP+FP+FN}$
ROC: FPR vs TPR=Recall
PR: Recall vs Precision

Maximum A Posteriori (MAP)

$\mathbf{w}^* = \arg\max_\mathbf{w}\ p(\mathbf{w}|\mathcal{D}) = \arg\max_\mathbf{w}\ p(\mathcal{D}|\mathbf{w})\,p(\mathbf{w})$
$\ell_2 \to p(\mathbf{w}) = \mathcal{N}(\mathbf{w}|\mathbf{0},\lambda^{-1}\mathbf{I}) \to \mathcal{L} + \lambda\|\mathbf{w}\|^2$
$\ell_1 \to p(\mathbf{w}) = \text{Laplace}(\mathbf{w}|\mathbf{0},\lambda^{-1}) \to \mathcal{L} + \lambda\|\mathbf{w}\|_1$

Bayes

$p(\mathbf{w}|\mathcal{D}) = \frac{p(\mathcal{D}|\mathbf{w})\,p(\mathbf{w})}{p(\mathcal{D})} \propto p(\mathcal{D}|\mathbf{w})\,p(\mathbf{w})$
posterior $\propto$ *likelihood* $\times$ *prior*

Distributions + MLE estimators

Gaussiana $\mathcal{N}(x|\mu,\sigma^2)$:
$$\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$
$\hat{\mu} = \frac{1}{N}\sum_i x_i$ $\hat{\sigma}^2 = \frac{1}{N}\sum_i(x_i-\hat{\mu})^2$
Gaussiana Multivariada $\mathcal{N}(\mathbf{x}|\boldsymbol{\mu},\boldsymbol{\Sigma})$:
$$\frac{1}{(2\pi)^{d/2}|\boldsymbol{\Sigma}|^{1/2}}\exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right)$$
$\hat{\boldsymbol{\mu}} = \frac{1}{N}\sum_i\mathbf{x}_i$ $\hat{\boldsymbol{\Sigma}} = \frac{1}{N}\sum_i(\mathbf{x}_i-\hat{\boldsymbol{\mu}})(\mathbf{x}_i-\hat{\boldsymbol{\mu}})^\top$
Bernoulli $\text{Ber}(x|\mu)$:
$$\mu^x(1-\mu)^{1-x}$$
$\hat{\mu} = \frac{1}{N}\sum_i x_i$
Categórica $\text{Cat}(y|\boldsymbol{\pi})$:
$$\prod_c \pi_c^{\mathbb{I}[y=c]},\quad \sum_c \pi_c = 1$$
$\hat{\pi}_c = \frac{N_c}{N}$ (frecuencia de clase c)
Laplace $\text{Lap}(x|\mu,b)$:
$$\frac{1}{2b}\exp\left(-\frac{|x-\mu|}{b}\right)$$
$\hat{\mu} = \text{median}(\{x_i\})$ $\hat{b} = \frac{1}{N}\sum_i|x_i-\hat{\mu}|$
Binomial $\text{Binom}(k|n,p)$:
$$\binom{n}{k}p^k(1-p)^{n-k}$$
$\hat{p} = \frac{1}{N}\sum_i\frac{k_i}{n}$

Information Theory

Entropy: $\mathbb{H}(\mathbf{p}) = -\sum_i p_i\log p_i$
Cross-E: $\mathbb{H}_{ce}(\mathbf{p},\mathbf{q}) = -\sum_i p_i\log q_i$
Binary Cross-E:
$\mathbb{H}_{ce}(p,q) = -[p\log q + (1-p)\log(1-q)]$
Joint-E: $\mathbb{H}(X,Y) = -\sum p(x,y)\log p(x,y)$
Conditional-E: $\mathbb{H}(Y|X) = \mathbb{H}(X,Y) - \mathbb{H}(X)$
Relative-E (KL divergence)
$\mathbb{H}(p\|q) = \sum_i p_i\log\frac{p_i}{q_i} = \mathbb{H}(p\|q) - \mathbb{H}(p)$
Mutual Information
$I(X;Y) = \mathbb{H}(p(x,y)\,\|\,p(x)p(y))$
$I(X;Y) = 0 \Leftrightarrow X \perp Y$

Linear Algebra

SDP: $\mathbf{A} \succeq 0 \Leftrightarrow \mathbf{x}^\top\mathbf{A}\mathbf{x} \geq 0\ \forall\mathbf{x}$.
$\mathbf{A}^\top\mathbf{A}$, $\mathbf{A}\mathbf{A}^\top \succeq 0$ siempre
Cond: $\kappa(\mathbf{A}) = \sigma_{\max}/\sigma_{\min}$, $\kappa(\mathbf{A}^\top\mathbf{A}) = \kappa(\mathbf{A})^2$
Precisión: $\boldsymbol{\Lambda} \triangleq \boldsymbol{\Sigma}^{-1}$
Derivadas conocidas
- $\nabla_\mathbf{x}(\mathbf{x}^\top\mathbf{A}\mathbf{x}) = 2\mathbf{A}\mathbf{x}$ ($\mathbf{A}$ simétrica)
- $\nabla_\mathbf{x}(\mathbf{a}^\top\mathbf{x}) = \mathbf{a}$
- $\nabla_\boldsymbol{\Sigma}\log|\boldsymbol{\Sigma}| = \boldsymbol{\Sigma}^{-\top}$

Traza: $\mathbf{x}^\top\mathbf{A}\mathbf{x} = \text{tr}(\mathbf{A}\mathbf{x}\mathbf{x}^\top)$, $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$
Jacobiano: $\mathbf{J} = \frac{\partial\mathbf{f}}{\partial\mathbf{x}} \in \mathbb{R}^{m\times n}$, $J_{ij} = \frac{\partial f_i}{\partial x_j}$
Si $m=1 \implies \mathbf{J} = \nabla f^\top$.

Convexity

$f(\lambda\mathbf{x}+(1-\lambda)\mathbf{y}) \leq \lambda f(\mathbf{x}) + (1-\lambda)f(\mathbf{y})$
1er orden: $f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top(\mathbf{y}-\mathbf{x})$
2do orden: $\nabla^2 f(\mathbf{x}) \succeq 0\ \forall\mathbf{x}$ (Hessiano SDP)

Ensemble Methods

Bagging: bootstraping aggregating.
Boosting: entrenamiento secuencial, ponderar errores. f_m se entrena sobre residuos de f_{m-1}
Random Forest: Bagging + features bagging por split.